

FoodLens

AI-Powered Ingredient Scanner and Health Risk Scoring System

Database Table Design (Schema)

1. Overview

This document specifies the relational database schema for the FoodLens system, implemented in MySQL. The design is normalized to avoid data redundancy, with clear primary key (PK) and foreign key (FK) relationships linking users, health profiles, products, ingredients, scan history, and community-submitted data. Thirteen core tables are defined below, grouped by functional area.

2. User & Health Profile Tables

2.1 Users

Stores core account information for all registered users, including end users, community contributors, and administrators.

Field	Data Type	Key	Description
user_id	INT	PK	Unique identifier for each user, auto-incremented.
name	VARCHAR(100)		Full name of the user.
email	VARCHAR(150)	UNIQUE	User's email address, used for login and report delivery.
password_hash	VARCHAR(255)		Securely hashed user password.
role	ENUM('user','contributor','admin')		Defines the user's role and access level within the system.
created_at	DATETIME		Timestamp of account creation.

2.2 Health Profiles

Stores individual health profiles linked to a user account, supporting the family/shared profile feature where one account manages multiple profiles.

Field	Data Type	Key	Description
profile_id	INT	PK	Unique identifier for each health profile.
user_id	INT	FK → Users	References the account that owns/manages this profile.
profile_name	VARCHAR(100)		Display name for the profile (e.g., "Myself", "Amma", "Child").
relation	VARCHAR(50)		Relationship of the profile to the account holder (self, parent, child, other).
created_at	DATETIME		Timestamp of profile creation.

2.3 Health Conditions

Stores chronic health conditions associated with a specific health profile, used to drive personalized score multipliers.

Field	Data Type	Key	Description
condition_id	INT	PK	Unique identifier for each recorded condition.
profile_id	INT	FK → Health Profiles	References the health profile this condition belongs to.
condition_name	VARCHAR(100)		Name of the condition (e.g., Diabetes, Hypertension, Obesity).
severity	ENUM('mild','moderate','severe')		Optional severity level, usable for finer-grained score weighting.

2.4 Allergies

Stores allergen sensitivities associated with a health profile, used to trigger hard allergen warnings independent of the numeric score.

Field	Data Type	Key	Description
allergy_id	INT	PK	Unique identifier for each recorded allergy.
profile_id	INT	FK → Health Profiles	References the health profile this allergy belongs to.
allergen_name	VARCHAR(100)		Name of the allergen (e.g., Peanuts, Milk, Gluten).

3. Ingredient & Scoring Tables

3.1 Ingredients

The core ingredient risk database, storing each ingredient's category, base risk score, and citation source, and powering the searchable ingredient glossary.

Field	Data Type	Key	Description
ingredient_id	INT	PK	Unique identifier for each ingredient.
name	VARCHAR(150)		Canonical ingredient name (e.g., "Sodium Benzoate").
category	VARCHAR(50)		Ingredient category (sweetener, preservative, fat, flavor enhancer, color, natural, etc.).
base_risk_score	DECIMAL(3,1)		Base risk score on a 0–10 scale, prior to any personalization.
allergen_flag	BOOLEAN		Indicates whether this ingredient is a known common allergen.
source_reference	VARCHAR(255)		Citation for the risk score (e.g., FSSAI guideline, WHO/ICMR reference).
created_at	DATETIME		Timestamp the ingredient record was added.

3.2 Ingredient Aliases

Stores alternate names and additive codes for each ingredient (e.g., E-numbers), enabling reliable alias resolution during NLP parsing.

Field	Data Type	Key	Description
alias_id	INT	PK	Unique identifier for each alias record.
ingredient_id	INT	FK → Ingredients	References the canonical ingredient this alias maps to.
alias_name	VARCHAR(150)		Alternate name or additive code (e.g., "E211", "Sucrose").

3.3 Condition Multipliers

Stores the personalization rules that adjust ingredient category risk scores based on a user's health conditions.

Field	Data Type	Key	Description
multiplier_id	INT	PK	Unique identifier for each multiplier rule.
condition_name	VARCHAR(100)		Health condition this rule applies to (e.g., Diabetes, Hypertension).
ingredient_category	VARCHAR(50)		Ingredient category the multiplier is applied to (e.g., sweetener, sodium-containing).
multiplier_value	DECIMAL(3,2)		Multiplication factor applied to the base risk score (e.g., 1.50).

4. Product & Scan Tables

4.1 Products

Stores product-level information, whether sourced from the Open Food Facts API or verified through community submission.

Field	Data Type	Key	Description
product_id	INT	PK	Unique identifier for each product.
barcode	VARCHAR(50)	UNIQUE	Product barcode/QR value, where available.
name	VARCHAR(200)		Product name.
brand	VARCHAR(150)		Product brand or manufacturer.
source	ENUM('open_food_facts','community')		Indicates whether the product data originated from the API or community submission.
verified_status	ENUM('verified','pending')		Data quality status of the product record.
created_at	DATETIME		Timestamp the product record was added.

4.2 Product Ingredients

A mapping table linking products to their constituent ingredients, preserving each ingredient's position in the original ingredient list.

Field	Data Type	Key	Description
product_ingredient_id	INT	PK	Unique identifier for each product-ingredient mapping.
product_id	INT	FK → Products	References the associated product.

Field	Data Type	Key	Description
ingredient_id	INT	FK → Ingredients	References the associated ingredient.
list_position	INT		Position of the ingredient in the product's printed ingredient list, used for weighting.

4.3 Scan History

Records every product scan performed by a user, including the computed score, supporting consumption tracking and reporting features.

Field	Data Type	Key	Description
scan_id	INT	PK	Unique identifier for each scan event.
user_id	INT	FK → Users	References the user who performed the scan.
profile_id	INT	FK → Health Profiles	References the health profile the scan was scored against.
product_id	INT	FK → Products	References the scanned product.
computed_score	DECIMAL(5,2)		Final personalized health score computed for this scan.
scanned_at	DATETIME		Timestamp of the scan event.

5. Community & Administration Tables

5.1 Community Submissions

Stores ingredient data submitted by users for unrecognized regional products, pending verification before being merged into the main Products table.

Field	Data Type	Key	Description
submission_id	INT	PK	Unique identifier for each community submission.
product_name	VARCHAR(200)		Name of the submitted product.
submitted_ingredients	TEXT / JSON		Draft ingredient list submitted by the user, pre-filled via OCR.
submitted_by	INT	FK → Users	References the contributing user.
status	ENUM('pending','approved','rejected')		Current review status of the submission.
consensus_count	INT		Number of matching independent submissions received for this product.
submitted_at	DATETIME		Timestamp of submission.

5.2 Admin Reviews

Logs administrator decisions on community submissions, providing an audit trail for data quality moderation.

Field	Data Type	Key	Description
review_id	INT	PK	Unique identifier for each review action.
submission_id	INT	FK → Community Submissions	References the submission being reviewed.
admin_id	INT	FK → Users	References the administrator who performed the review.
decision	ENUM('approved','rejected')		Outcome of the admin review.
reviewed_at	DATETIME		Timestamp of the review decision.

5.3 Reports

Tracks generated exportable or scheduled reports (PDF or email) produced for users from their scan history.

Field	Data Type	Key	Description
report_id	INT	PK	Unique identifier for each generated report.
user_id	INT	FK → Users	References the user the report was generated for.
report_type	ENUM('pdf_export','scheduled_email')		Type of report generated.
generated_at	DATETIME		Timestamp the report was generated.

6. Entity Relationship Summary

Users (1) → (M) Health Profiles → (M) Health Conditions / Allergies. Ingredients (1) → (M) Ingredient Aliases. Products (M) ↔ (M) Ingredients via Product Ingredients. Users (1) → (M) Scan History (M) ← (1) Products, scored against a specific Health Profile. Users (1) → (M) Community Submissions (1) → (M) Admin Reviews. Users (1) → (M) Reports. Condition Multipliers are referenced at scoring time by matching a profile's Health Conditions against an ingredient's category, rather than through a direct foreign key.